

Project title: Designing algorithm visualizer for CPU scheduling algorithms

Task description:

Students are required to design and develop a software application such as a website, mobile application, or desktop application for the visualization and comparative analysis of CPU scheduling algorithms.

Module 1: Algorithm Visualization

This module must include the following components:

- A user interface for accepting input parameters from the user.
- Functionality allows the user to select a scheduling algorithm and specify the required inputs (number of processes, arrival time, burst time, and priority value where applicable) for each process.
- A visual representation of the corresponding Gantt chart based on the provided input data.
- A tabulated output displaying, for each process: completion time, turnaround time, and waiting time. The application must also compute and display the average waiting time, average turnaround time, and total CPU idle time.

Module 2: Comparative Analysis

This module must include the following components:

- A user interface for accepting input parameters from the user.
- The ability for the user to select two or more scheduling algorithms for comparison.
- Input collection for each of the selected algorithms.
- Gantt chart visualization for each selected algorithm.
- A comparative summary highlighting the performance differences among the selected algorithms.

Breakdown of marks

- Project report: 10
- Task accomplishment and implementation: 20
- Viva: 10